
When Can Language Models Join Without Lexical Cues?

Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity

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Abstract

We ask whether 2-hop answering over small labeled graphs needs readable relation strings, or only equality among identifiers. **Opaque isomorphic reasoning (OIR)** is a measurement protocol: a renaming σ of entities and relations that we verify to be injective on each instance’s finite atom set, with no legend in the prompt. HMAC-SHA256 is one generator of σ , not a cryptographic or privacy primitive. We score the answer node, not an emitted (r_1, r_2) .

Finding. Unique sealed routes still join (OpenAI unique English/opaque 200/200 on a city-typed $n=200$ freeze; opaque unique 32/32 with the start in q ; both-opaque unique 32/32 on three families). Under two competing sealed routes, **named-arm** OpenAI (readable case ids plus an ARM line that names two 2-hops) is 38/200 (UNKNOWN 151, decoy 11). The same graphs with hashed ids, ARM line kept, are 134/200 (coverage 49/200 $\rightarrow$ 155/200). With hashed ids and no ARM, at 4096/medium, English two-path is 197/200 and opaque two-path is 174/200. English relation names restore OpenAI with anonymous entities (opaque-entity two-path 32/32 vs. both-opaque 1/32). On the primary 2×2 , seals hide relation strings; entity names remain. They are not a confidentiality proof.

1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks `works_at` then `headquartered_in` to Burbank. That success may use English predicates, or only equality among identifiers. The two-path decoy is an injected second person $\rightarrow$ company $\rightarrow$ city route (`partner_of/located_in`) to Kansas City, not a Wikidata fact. Opaque two-path keeps the English question and entity names; those predicates become HMAC tokens (`works_at` $\mapsto$ E334675310f05 on the scored Eisner item).

Opaque isomorphic reasoning (OIR) measures whether a model can join after English predicate strings are replaced by keyed opaque tokens, with no legend in the prompt. A 2-hop factors as two inferred maps; we score only the answer node: selection $q \mapsto (r_1, r_2)$ and traversal $(v_0, r_1, r_2, G) \mapsto v_2$. Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in q as the same atom as in V (**start in q**) separates that question from start-entity identification. On the primary 2×2 , “lexical cues” means relation strings (entities stay English); both-opaque is a separate ablation. That suite is WIKI-H5. **Named-arm** means a one-line condition header plus readable case ids (F22_OPAQUE_AMBIG_1); hashed-id cells replace those ids with hashes.

Figure 1 plots the $n=32$ discovery beside the city-typed $n=200$ freeze (unique-path 200/200, named-arm opaque two-path 38/200). This paper reports three measurements. (i) Unique sealed routes still join with the start in q and no written hop. (ii) Under two competing sealed routes the usual *named-arm* reply is UNKNOWN, not the decoy; exact-match still moves with prompt surface, including a no-ARM hashed-id English/opaque pair of 197/200 vs. 174/200 (Section 5.2). (iii) Matched demonstrations restore shown seals; novel seals do not (a control whose expected gold is near zero). What is new is how opacity meets path ambiguity under a named-arm prompt (Section 5.1), and how large a prompt-surface move can be on the same sealed graph (Section 5.2).

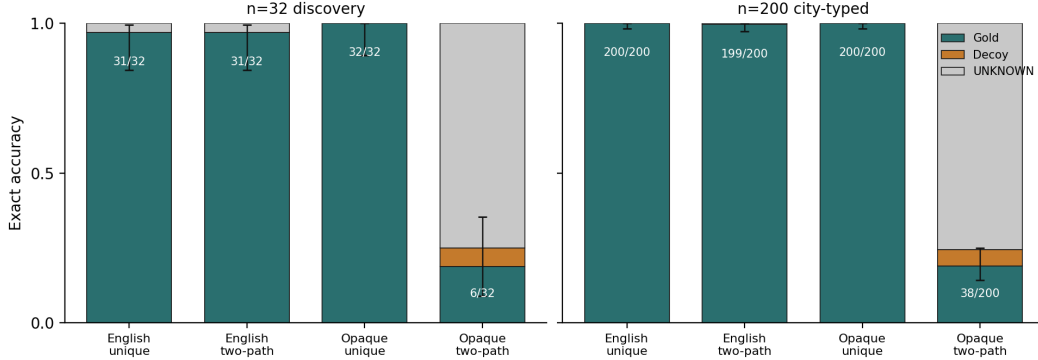


Figure 1: OpenAI WIKI-H5 2×2 (start in q). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals. Left: $n=32$ discovery (August lock; opaque two-path 6/32). Right: city-typed $n=200$ freeze (named-arm opaque two-path 38/200, UNKNOWN 151, decoy 11). Hashed-id 134/200 and alias $k=3$ are Section 5.2.

2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next readable relation (Sun et al., 2024; Luo et al., 2024). Ciphered ICL is not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps entity names in English math and templates some numerical values (Mirzadeh et al., 2025); 2WikiMultihopQA is a readable multi-hop control (Ho et al., 2020).

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo HMAC-seals entities (ϕ_Q , e.g. `ent_17`); relations stay original labels or clustered schema words, not keyed opaque tokens, and a local Hand answers on the raw graph after a remote Brain ranks retrieved paths (CWQ 67% $\rightarrow$ 59%; WebQSP 75%) (Tan et al., 2026). CoR explores readable relations because entities often have no semantics beyond opaque IDs — CoR’s opaque-entity hide-set, not relation-only WIKI-H5 (Liu et al., 2026). Tang et al. (2023) find LLMs fail when relation names are replaced by symbols (`r1`) and rules must carry deduction; OpenAI unique-path opaque 32/32 is the complementary case: sealed relations still traverse when topology fixes the route. Guo et al. (2025) report collapse to random guessing among end entities when distractor two-hops have other sources. Named-arm WIKI-H5 opaque two-path, with UNKNOWN allowed, is abstention rather than guessing (Section 5.2). A forced-choice arm is not in this evaluation (Section 7) and would put those protocols on the same scale. OIR withholds retrieved paths on the free arms (Table 1).

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound. NLGraph and Talk like a Graph encode graphs without readable typed predicates (Wang et al., 2023; Fatemi et al., 2024); OIR starts from a labeled KG and removes only the relation lexicon. Case-id strings that name the condition (OPAQUE_AMBIG) are semantic content, not only a meaning-preserving format perturbation in the sense of Sclar et al. (2024); we still treat packing as a factor. Symbol tuning, binding-ID analyses, and compositional-limit studies change the labels or the task (Wei et al., 2023; Feng and Steinhardt, 2024; Dziri et al., 2023). OIR holds the join, hop length, graph size, and question template fixed and replaces relation strings; OpenAI unique-path opaque traversal stays

Work	Hidden	Remote task	Join under opacity
ICL	substitution text	ICL task accuracy (bijection inferred)	ICL tasks
CIPHERS			
ARoG	entities; relations readable; topic names in q	retrieve + generate (relations, concepts)	privacy KG-RAG
PrivGemo	entities (HMAC); relations original or clustered schema words	rank retrieved paths; local plaintext answer	privacy KG-RAG
CoR	opaque entity IDs; relations readable	relation-centric exploration	KGQA
OIR	relations (primary 2×2); entities too (ablation); one join or none	emit graph-namespaces answer (no local decode)	selection vs. traversal

Table 1: Related protocols. PrivGemo HMAC-seals entities; relations remain original or clustered schema labels, and a local Hand answers on retrieved paths (CWQ/WebQSP). OIR withholds those paths on the free arms and scores the answer token from the same model; 38/200 is not a CWQ rate.

at 32/32, so that removal withdraws the cue that would otherwise select between routes rather than blocking execution.

3 Problem Formulation

Let $G = (V, R, T)$ be a labeled multi-relational graph ($T \subseteq V \times R \times V$) and q a natural-language or programmatic query with answer $a \in V$. A **seal** σ is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 is one generator of σ , not a cryptographic bound (Appendix A). The model receives G or $\sigma(G)$ and a query form built from σ (or from English, depending on the arm) and must return the answer node in that namespace; recovering σ^{-1} is not scored.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

Traversal is set-valued ($\tau_G: V \times R \times R \rightarrow 2^V$). We evaluate only instances for which $|\tau_G(v_0, r_1, r_2)| = 1$ for the gold plan (and, on two-path items, for the decoy plan), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in G .” A sealed join plan supplies (r_1, r_2) and tests (2). The CEO free-English arm hashes q piecewise, so v_0 typically does not appear as a single graph atom; it is not a pure test of (1).

A 2-hop is *structurally executable* when (r_1, r_2) is given, or when topology leaves only one 2-hop from a start already in q . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates); we do not claim mathematical unidentifiability. Any gold on that arm must use residual prompt cues, parametric memory of the named entities, or other unmeasured information; we do not observe (r_1, r_2) . Low exact-match is not evidence that traversal failed. Dual-path separates the cases: shown seals restore end-to-end accuracy; novel seals do not. The scored output is the answer node or UNKNOWN, not (r_1, r_2) . We infer whether selection was supported from those outcomes and from controls (unique topology, written hop, matched vs. novel seals). Throughout, “selection failure” names that inference about the latent (r_1, r_2) stage; we do not observe the pair. Models may abstain (UNKNOWN) rather than guess (selective prediction; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). Coverage is the fraction of items with a non-UNKNOWN parse (other and both-named count; empty completions stored as UNKNOWN do not; NO_OUTPUT

does not); selective accuracy is exact match among those items, not a forced-choice among the two routes (Kadavath et al., 2022; Wen et al., 2025). NO_OUTPUT is an empty completion on a runner that does not store empties as UNKNOWN (Wiki-H5/Wiki-OO). **Both-named** is a parse that emits both person tails.

After sealing, the Eisner unique-path CONTEXT is Michael_Eisner | E334675310f05 | The_Walt_Disney_Company then The_Walt_Disney_Company | Eac04fab538ca | Burbank (12-hex tokens; entities remain English on the primary 2×2).

We use five measurement arms. **Readable English** is the baseline (start in q ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map. **Sealed join plan** writes (r_1, r_2) over σ -atoms (execution control); **SealRouter** is the non-LLM exact joiner on that sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in G typically does not appear in q (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

4 Method

Seal recipe. Atoms are normalized then HMAC-SHA256-renamed with a per-item 16-byte key (token E plus 12 hex digits). HMAC is keyed renaming, not a security mechanism; 48-bit truncation is instance-wise injective on these small atom sets (normalization rules, type non-separation, and collision checks: Appendix A). The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess that latter variation (Table 8). Hashing q token-by-token does not reconstruct the graph atom Michael_Eisner; that is the missing-start failure. Reported opaque cells are this HMAC construction, not every opaque identifier distribution.

Matched 2×2 (start in q). The $n=32$ suite is a seeded random sample of 32 of 73 Wikidata-derived CEO records (random.Random(20260821).sample; Vrandečić and Krötzsch, 2014), not a published CEO benchmark and not a hand-picked subset. Each instance is 4 triples (unique-path: 6 entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses works_at then the headquarters relation; the question says “led by”. The two-path condition *injects* a competing route: ambig_edges attaches record $i+1$ ’s company/HQ under partner_of/located_in. That decoy is not a Wikidata fact. It isolates same-type competition on a fixed template, not naturally occurring CWQ/WebQSP ambiguity. To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length, template, and graph size fixed. Two factors vary: relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Tails are intended as the same type; Wikidata P159 was not filtered (buildings, stations, a street, and one state among gold HQs). Both OpenAI opaque two-path decoys sit on non-city golds with a city decoy tail (Appendix A). On Wikidata two-path graphs both intermediates are companies (works_at/headquartered_in vs. partner_of/located_in). In the primary run the gold 2-hop is listed before the decoy 2-hop in CONTEXT (same order in English and opaque). Listing-order robustness is Section 5.2: a new draw on the $n=200$ freeze, not a reordering of the primary completions. The $n=32$ shuffle is a later snapshot (factorial_2x2_iso_shuffle_gpt56.json), not that comparison.

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only.
No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

It does *not* contain the equality-recipe line (Appendix J). The format token is <city_or_UNKNOWN>, but six $n=32$ gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit UNKNOWN. The powered city-typed cell is $n=200$

(Section 5.1); the $n=32$ lock is the three-family pin. OpenAI English two-path is 31/32 under that CONTEXT-only instruction, so the line does not by itself force abstention. Readable names (Eisner, Burbank) still leave OpenAI at UNKNOWN 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

Further constructions. The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in q . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (Wiki-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only 2×2 . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). MOVIE-A32 and MOVIE-A100 leave English verbs such as *directed* in q ; MOVIE-B hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds G fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

Scoring. The format asks for one answer token in the graph’s namespace or UNKNOWN (plain-text city when entities stay English; $\sigma(a)$ when entities are sealed). Primary metric: exact match (UNKNOWN is a miss). Scored UNKNOWN on Wiki-H5/Wiki-OO is an explicit UNKNOWN line. Dual and some no-hop OpenAI cells stored empty completions as UNKNOWN. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use $z=1.96$ (Wilson, 1927) ($0/32 \Rightarrow [0.00, 0.11]$; $6/32 \Rightarrow [0.09, 0.35]$; $32/32 \Rightarrow [0.89, 1.00]$). Those intervals treat the 32 items as Bernoulli trials on a single draw; they are not a generation-variance interval. OpenAI Wiki-H5 $k=3$ at $n=32$ and $n=200$ is Section 5.1. On the OpenAI Wikidata 2×2 , English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided complete-case $p=5.96 \times 10^{-8}$; McNemar, 1947). That is the two-path paired contrast, not an interaction test, and it largely measures removing the cue that made one sealed route identifiable. The empirical content is the response mode (UNKNOWN vs. decoy) and prompt-surface packing (Section 5.2). **Isolation** means one quiz per file with gold answers stored outside the model-visible prompt (stricter than multi-quiz **batch** files). Isolation $n=32$ uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros.

Models. The primary family is OpenAI gpt-5.6-sol (Chat Completions; unpinned alias; temperature omitted; 512 completion tokens unless noted). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK on the same quiz files, with a reply-only preamble OpenAI never received. Unique-path opaque still executes on those SDK cells (Composer 27/32, OpenAI 32/32, Grok 30/32; Table 7). Tools, retry, and empty-reply handling are not equated. The August lock stores the parsed answer, not the API model string. Inference settings and $k=3$ alias draws are Appendix A. A packaging-only Claude probe is Appendix F. Isolation quizzes and locked scores are at <https://github.com/pankajmits/oir>.

5 Results

Table 2 names the cells. OpenAI is primary; Composer/Grok used a different harness (Table 7). MOVIE-B is a cross-dataset check, not a Wiki-H5 resample. Packaging probes, plus Spider $n=100$ and 2Wiki-CF $n=200$, are Appendix G.

5.1 Opacity $\times$ path ambiguity

Wiki-H5 holds the start string in q and varies only relation lexicality (English vs. HMAC) and the number of 2-hops from that start (one vs. two). OpenAI is the primary family. Composer 2.5 and Grok 4.5 used Cursor’s agent SDK (Table 7).

ID	Dataset	Condition	n	Models	Scored
WIKI-H5	Wikidata CEO	rel-only 2×2 , start in q	32	OpenAI (primary)	node
WIKI-H5-200	Wikidata CEO, city P31	same 2×2 , QID freeze	200	OpenAI	node
Header $\times$ decoy	same 32 people	ARM line $\times$ decoy, hashed ids	32	3 families	node
Header $\times$ decoy-200	same 200 people	ARM $\times$ decoy, hashed ids	200	OpenAI	node
WIKI-OO	same graphs	both-opaque	32	3 families	$\sigma(a)$
WIKI-OO-200	$n=200$ freeze	both-opaque	200	OpenAI	$\sigma(a)$
WIKI-SHUF	same two-path	listing order	32	OpenAI	city
WIKI-SHUF-200	$n=200$ freeze	listing order	200	OpenAI	city
MOVIE-A32	WikiMovies	actor $\rightarrow$ movie $\rightarrow$ dir vs writer	32	3 models	person
MOVIE-A100	WikiMovies	same hops; verbs in q	100	OpenAI; Comp./Grok A	person
MOVIE-A200	WikiMovies	$n=100$ prefix + 100 new actors	200	OpenAI	person
MOVIE-B	WikiMovies	same hops; verbs hashed	100	OpenAI	person
MOVIE-B200	WikiMovies	Condition B on $n=200$ freeze	200	OpenAI	person
DUAL	WikiMovies/synth	matched vs. novel demos	32	3 models	$\sigma(a)$
DUAL-200	WikiMovies $n=200$	matched vs. novel	200	OpenAI	$\sigma(a)$
CEO-NL	Wikidata CEO	piecewise q , no plan	32	3 models	$\sigma(a)$
CEO-NL-200	$n=200$ freeze	piecewise q , no plan	200	OpenAI	$\sigma(a)$

Table 2: Cell inventory. WIKI-H5-200 is the powered city-typed freeze (OpenAI). Composer/Grok $n=32$ SDK cells; Table 7. Six $n=32$ P159 golds are not cities (Appendix A). A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC.

City-typed $n=200$ (OpenAI). The $n=32$ source file has 73 CEO rows and six non-city P159 golds. WIKI-H5-200 is a new freeze: 200 unique people and companies, HQ frequency capped at 3, every HQ an instance of a city-like Wikidata class (P31 list in Appendix A), person/company/HQ QIDs stored, and all 200 P169+P159+P31 triples re-checked live against the Query Service (`wikidata_ceo_hops_n200.json`). It does not overwrite the $n=32$ lock. OpenAI named-arm: unique English and unique opaque 200/200, English two-path 199/200, opaque two-path 38/200 (UNKNOWN 151, decoy 11; Wilson [0.14, 0.25]), plan 200/200. The 38/200 rate sits inside the $n=32$ Wilson band for 6/32. Decoy 0 on the $n=26$ city slice was a small-sample artifact: $n=26$ cannot resolve a $\sim 5\%$ decoy rate. At $n=200$ the usual named-arm reply is still UNKNOWN ($\approx 14:1$ vs. decoy). This is not a CWQ/WebQSP number: those protocols retrieve paths and answer locally; OIR withholds retrieved paths on the free arms (Table 1). Prompt-surface packing of this freeze is Section 5.2. $n=200$ is OpenAI-only. Headline locks are `gpt56n200 / gpt56n200h512`; alias $k=3$ of those two 512-token cells is below. Three-family cells stay at $n=32$.

$n=32$ discovery. On unique sealed routes OpenAI executes (opaque unique 32/32; plan 32/32). On competing sealed routes the usual named-arm reply is UNKNOWN, not the decoy. On 26 city-valued items unique English, unique opaque, and English two-path are 26/26; named-arm opaque two-path is 5/26, 2/26, 2/26 across three draws. August: UNKNOWN 21, decoy 0, selective accuracy 5/5. The lock is opaque two-path 6/32 (UNKNOWN 24, decoy 2); both decoys sit on non-city P159 golds. Wilson: $6/32 \Rightarrow [0.09, 0.35]$; city slice $5/26 \Rightarrow [0.09, 0.38]$. The left panel of Figure 1 is this $n=32$ lock.

Let Δ_{opq} be opaque minus English exact-match on $n=32$: unique $+1/32$, two-path $-25/32$ (25 English-correct and opaque-wrong, 0 reverse; McNemar $p=5.96 \times 10^{-8}$). Opacity is not a randomized within-item treatment (Appendix I).

OpenAI $k=3$. Alias draws of the unpinned snapshot are not a generation-variance interval. At $n=32$ (15 September; `gpt56k3r2/r3`): named-arm 6/32, 3/32, 3/32; hashed-id H5-cyclic 20, 21, 22; Composer 10, 7, 6 and Grok 1, 0, 0. OpenAI named-arm draws 2–3 are both 3/32 and Grok both 0/32; Composer is 7 then 6. Hashed-id H5-cyclic has no August draw. August named-arm is highest in all three families. At $n=200$ (16 September; `gpt56n200k3r2/r3` and `gpt56n200h512k3r2/r3`): named-arm 38/200, 46/200, 43/200 (UNKNOWN 151, 144, 142; decoy 11, 10, 15); hashed-id 134, 130, 127 (no NO_OUTPUT on draws 2–3). Those tags do not replace `gpt56n200/gpt56n200h512`. The two cells do not meet (gaps 96, 84, 84). Coverage is 49/56/58 vs. 155/153/145; selective 38/49, 46/56, 43/58 vs. 134/155, 130/153, 127/145. Gold on all three draws is 28 vs. 97. On OpenAI the other $n=32$ named-arm cells hold: unique

opaque 32/32, 31/32, 32/32; English two-path 31/32, 30/32, 30/32; plan 32/32 on every draw. Named-arm English two-path is OpenAI 31/32, Composer 26/32, Grok 4/32: a packaging $\times$ family effect, not a ranking. Opaque unique stays high (Composer 27/32, Grok 30/32); a written hop recovers (28/32, 32/32). Two-path coverage / selective accuracy (named-arm): OpenAI opaque 8/32, 6/8; Composer 10/32, 10/10; Grok 1/32, 1/1. Two other generators still have unique > two-path (Table 8).

OpenAI Wiki-H5 (gold / decoy / UNK)	Named $n=32$	Named $n=200$	Hashed $n=200$
English unique	31/0/1	200/0/0	—
English two-path	31/0/1	199/0/1	—
Opaque unique	32/0/0	200/0/0	—
Opaque two-path	6/2/24	38/11/151	134/20/44

Table 3: OpenAI response modes. Named-arm uses readable case ids. Hashed-id 512 keeps the ARM line (remainder: other 1, NO_OUTPUT 1). Coverage 49/200 $\rightarrow$ 155/200; selective accuracy 38/49 $\rightarrow$ 134/155. Alias $k=3$ gold 38/46/43 vs. 134/130/127; coverage 49/56/58 vs. 155/153/145. Same-date $n=32$ hashed-id is 20/1/8 (other 2, NO_OUTPUT 1). Composer/Grok SDK cells: Table 7.

5.2 Prompt surface as a factor

The named-arm WIKI-H5 prompt contains this line:

```
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

That sentence tells the model in English that two 2-hops exist. It is an uncontrolled treatment if abstention is attributed only to opacity-meets-ambiguity. Prompt-surface packing is a factor (Sclar et al., 2024). Two manipulations must not be collapsed.

Case-id wording, ARM line held fixed. The case id appears in `Format: ANSWER_PLAIN[id]` and in the ID line. Named-arm uses readable ids (`F22_OPAQUE_AMBIG_i`); the H5-cyclic arm substitutes hashes. ARM line and CONTEXT are identical. On the city-typed $n=200$ freeze, the same 512-token contrast is 38/200 vs. 134/200 (+96 golds; H5-cyclic has one NO_OUTPUT). Coverage moves 49/200 $\rightarrow$ 155/200 and selective accuracy 38/49 $\rightarrow$ 134/155: that is not only more guessing. Decoy also rises 11 $\rightarrow$ 20. Headline $n=200$ is one lock each (gpt56n200, gpt56n200h512). Alias $k=3$ at $n=200$ is 38/46/43 vs. 134/130/127: the low cell moves by eight items, the high cell by seven, and they do not meet (gaps 96, 84, 84). Coverage stays 49/56/58 vs. 155/153/145. The $n=32$ pair is 6/3/3 vs. 20/21/22. The $n=32$ discovery pair is same-date 11 September: 5/32 vs. 20/32 (+15; one NO_OUTPUT each). Both prompts keep the ARM line, so this is not a test of whether the prompt announces ambiguity. $k=3$ hashed-id gold is 20, 21, 22; against named-arm 6/3/3 only +18 and +19 are same-day.

ARM line, case ids held hashed. Cyclic decoys attach record $i+1$'s company/HQ, as in WIKI-H5. Constant decoys reuse the first eligible pool record (`curve_decoy`). On $n=32$ that record is Cogent Communications $\rightarrow$ Washington, DC on every item. On the $n=200$ freeze it is San_Jose on 199/200 items and London on 1; gold=decoy is 0; Cogent is not that decoy. Table 4 crosses the ARM line with cyclic vs. constant decoys at 4096/medium on the $n=200$ freeze. H5 cyclic vs. constant McNemar 95 vs. 10, $p=1.6\times 10^{-18}$; no-ARM 30 vs. 7, $p=1.9\times 10^{-4}$. The $n=32$ discovery (Table 9) already showed a simple-effect gap under H5 (cyclic 20 vs. constant 10; McNemar 14 vs. 4, $p=0.031$) and not under no-ARM (gold 25 vs. 27; McNemar 1 vs. 3, $p=0.625$). At $n=200$ both H5 and no-ARM cyclic-vs-constant simple effects are significant; we do not report an item-level interaction test. K2 is a shorter ARM that still names two hops. K2 and no-ARM were not run at 512.

Listing order. Within one shuffle run on the $n=200$ freeze, gold-first is 25/100 and decoy-first is 34/100 (two-proportion $p\approx 0.16$, not significant). That is the listing test: same draw, same case-id family. Overall 59/200 versus the primary gold-first cell 38/200 is a new draw (case ids `F200S_` rather than `F200_`), not a within-run reordering effect; we do not read +21 as a listing-order gain.

The $n=32$ shuffle is a later snapshot (`factorial_2x2_iso_shuffle_gpt56.json`) and was too small to read a decoy-first effect (decoy 0/16 vs. 1/16).

ARM line (OpenAI, hashed ids, 4096/medium, $n=200$)	Cyclic	Constant
no ARM	174/200	151/200
H5	129/200	44/200
K2	179/200	144/200

Table 4: Header×decoy on the city-typed $n=200$ freeze (same people as WIKI-H5-200). Cyclic = record $i+1$ ’s HQ; constant = `curve_decoy` ($n=200$: San_Jose ×199, London ×1; not Cogent→DC). English, no ARM, cyclic is 197/200. H5 cyclic vs. constant McNemar 95 vs. 10 ($p=1.6\times 10^{-18}$). H5 vs. no-ARM on constant decoys is McNemar 4 vs. 111. Same-date 512-token hashed-id H5-cyclic is 134/200 (one NO_OUTPUT). $n=32$ golds: Table 9.

English Grok. Named-arm English two-path is Grok 4/32 (UNKNOWN 28). The English header×decoy cell (ENG_NONE_CYC) is 24/32. That cell has *no* ARM line and hashed ids, so two things differ from the 4/32 named-arm English prompt. Call it English, no ARM, hashed ids—not “arm-neutral English.” A prompt-verbatim Grok rerun of the named-arm English files (`grok45pn`) stays 4/32, so the Cursor preamble is not that gap. OpenAI named-arm English is 31/32 vs. English, no ARM, hashed ids 29/32: OpenAI English does not abstain either way.

What the data support: exact-match under competing sealed routes is strongly sensitive to prompt surface form. Case-id wording and the ARM line are large, non-exchangeable moves on the same sealed graphs (paragraphs above; Table 4). Budgets differ, so these are not a ranking of manipulations; the ARM effect is not a small add-on to case-id.

A consequence for the contrast with Guo et al. (2025). That work reports collapse to guessing among end entities when distractor two-hops have other sources. Our named-arm OpenAI cell is abstention (UNKNOWN 24/32, decoy 2/32 at $n=32$; UNKNOWN 151, decoy 11 at $n=200$), not a coin-flip onto the decoy. That “UNKNOWN rather than decoy” pattern is not a three-family law. Grok no-ARM opaque commits are 6 gold / 3 decoy cyclic and 1 / 5 constant (7 vs. 8 pooled): chance. For Grok, preferring UNKNOWN over the decoy is a named-arm phenomenon.

Composer/Grok hashed-id H5-cyclic cells used `--preamble none` and score 6/32 and 0/32; the named-arm locks used the Cursor preamble and score 10/32 and 1/32. Both families go *down* with hashed ids, a sign reversal relative to OpenAI. We did not re-query Composer named-arm with the prompt file verbatim, so that reversal mixes case-id wording with preamble deletion and is not interpretable. Grok is at floor on both opaque two-path cells.

5.3 Both-opaque and readable relations

Table 5 is a second isolation suite on the same 32 records (ANSWER_SEALED; start seal in q). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique (27/32, 30/32 in Table 7) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (Wilson [0.01, 0.16]), Composer 0/32, Grok 0/32. A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32 (UNKNOWN 29, decoy 0). On the $n=200$ freeze (OpenAI), unique both-opaque and OE two-path are 200/200; both-opaque two-path is 12/200.

5.4 WikiMovies: hashed verbs

Table 6 uses the same actor→movie→director vs. writer template as MOVIE-A32 (person-valued tails). MOVIE-A32 unique-path with English verbs in q is 32/32 for all three families; opaque two-path is 17/32, 17/32, 0/32, with plan 32/32. A32’s ARM names two 2-hops; A100’s is OPAQUE_AMBIG $n=100$. On 32 A32 items matched by gold name (31 unique names) OpenAI is 17/32 vs. 29/32 under A100’s thinner ARM. The extra 68 items are 67/68, which is why the $n=100$ cell is 96/100. Keys differ; they cannot explain an English-name lookup. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Condition A leaves

Arm	Ent.	Rel.	OpenAI	Composer	Grok
Unique path	opaque	Eng.	32/32	32/32	32/32
Two paths	opaque	Eng.	32/32	28/32	3/32
Unique path	opaque	opaque	32/32	32/32	32/32
Two paths	opaque	opaque	1/32	0/32	0/32
Two paths + plan	opaque	opaque	32/32	28/32	32/32

Table 5: WIKI-OO and opaque-entity (OE) on the same 32 graphs (start in q). OpenAI OE two-path 32/32; both-opaque two-path 1/32. Grok OE two-path 3/32 (UNKNOWN 29). Plan is a control.

WikiMovies people (OpenAI, $n=100$)	Unique	Two-path
English relations, Condition A (verb in q)	100/100	100/100
Opaque relations, Condition A (verb in q)	100/100	96/100
Opaque relations, Condition B (verbs hashed)	100/100	3/100
Opaque two-path + plan, Condition B	—	100/100

Table 6: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (OpenAI $n=100$, start in q). Condition B: unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; coverage 13/100, selective accuracy 3/13).

directed/starred in q ; we do not treat it as a selection result. Condition B hashes those verbs: OpenAI unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; Wilson [0.01, 0.08]). Coverage is 13/100 and selective accuracy 3/13. A written hop is 100/100. Composer/Grok Condition A two-path: 62/100, 36/100; Condition B $n=100$ is OpenAI-only. On the $n=200$ freeze, Condition A two-path is 122/200 (Wiki-H5 topology ARM, not A100’s ARM OPAQUE_AMBIG $n=100$ banner, so not a resample of 96/100); Condition B is 7/200. Movie-B’s ARM names two 2-hops (Wiki-H5-style); A100 Condition A does not. The contrast is a header-width comparison on WikiMovies, not a same-prompt Wiki-H5 replication (Section 5.2). The $n=32$ MetaQA-style template cell (Appendix H) is not MOVIE-B.

5.5 In-prompt seal reuse

Dual-path holds G fixed with two complete 2-hop routes and varies whether demonstrations expose the gold relation seals. Matched-relation is in-prompt reuse of shown correspondences, not induction of a general cipher. Novel-relation demos teach seals that do not appear on the gold route. An explicit join plan is the execution upper bound. Asymmetric vs. balanced noise: Appendix D.

Isolation $n=32$ (Figure 2, Appendix D): OpenAI saturates matched-relation under both noise regimes (32/32, decoy 0). Novel-relation balanced is exact match 0/32: all 32 replies are stored UNKNOWN (24 empty completions, 8 explicit UNKNOWN). The novel arm’s expected gold is near zero. Novel-relation asymmetric is also 0/32 (UNKNOWN 24, of which 17 empty, decoy 8). An explicit dual-path plan scores 32/32. WikiMovies director/writer Dual is 32/32 vs. 0/32 for all three families (dualpath_wikimovies_iso_*; Composer novel decoy 2/32), with plan 32/32. On the synthetic Dual suite, Composer isolation-tag novel-relation UNKNOWN (composer25) is a sealed-demo helper; free-form composer25ff is decoy 32/32. Grok free-form (grok45ff) novel-relation asymmetric is UNKNOWN on all 32 items. Last-demo answer-token copy is 0/32 on every arm. On the $n=200$ freeze (OpenAI), matched is 200/200 and novel is 2/200 (UNKNOWN 198, decoy 0), with plan 200/200.

On a two-path graph a forced uniform guess among the two complete routes is 1/2; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism. Arm definitions: Appendix D.

5.6 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5,

OpenAI, and Grok 4.5 (Table 11). On the $n=200$ freeze, plan remains 200/200 and free sealed English 0/200. We exclude a batched free-sealed cell that can succeed by returning a unique neighbor without using q .

6 Discussion

Sealed unique routes are executable: opaque unique 32/32 and 200/200, plan 32/32 and 200/200. Under two competing sealed routes, named-arm OpenAI usually abstains (38/200, UNKNOWN 151, decoy 11). Hashed case ids on the same graphs recover joins rather than guesses (coverage 49/200 $\rightarrow$ 155/200, selective 38/49 $\rightarrow$ 134/155). With hashed ids and no ARM, English two-path is 197/200 and opaque two-path is 174/200 (Table 4). Alias $k=3$ is 38/46/43 vs. 134/130/127 (coverage 49/56/58 vs. 155/153/145). That surface move is large; it does not make abstention the wrong description of the *named-arm* cell. Grok no-ARM opaque commits are near chance, so “UNKNOWN rather than decoy” is not a three-family law. English predicates restore OpenAI with anonymous nodes (32/32 vs. both-opaque 1/32); Grok OE two-path is 3/32 (UNKNOWN 29). Dual-path reuses shown seals (matched 200/200) and does not transfer to novel seals (2/200; expected gold near zero). A high English two-path rate does not imply the same join once relations are opaque. OIR withholds retrieved paths on the free arms; PrivGemo’s CWQ/WebQSP numbers keep those paths and answer locally (Tan et al., 2026). We do not claim mathematical unidentifiability. Hypothesis status: Table 10.

7 Limitations

OpenAI is the primary family. Headline $n=200$ is the `gpt56n200/gpt56n200h512` lock; alias $k=3$ of those 512-token cells is 38/46/43 vs. 134/130/127 (Section 5.1). $n=32$ $k=3$ is named-arm 6/3/3, hashed-id 20/21/22, unique opaque 32/31/32. Composer/Grok stay at $n=32$ on a different harness (Table 7). The competing two-path route is injected, not a Wikidata fact; graphs are four triples, smaller than retrieved KG-RAG subgraphs. The $n=26$ decoy-0 slice is underpowered for a $\sim 5\%$ decoy rate (Section 5.1). The $n=32$ format token makes six P159 golds ill-posed; $n=200$ does not use those golds. The $n=32$ suite is 32 of 73 records, one template, one seed; decoys are cyclic, so item-independence for Wilson intervals is slightly optimistic (McNemar uses paired items but still assumes independent pairs). $k=3$ at $n=32$ mixes the August lock with later alias draws (Section 5.1): OpenAI named-arm 3=3 and Grok 0=0, Composer 7 $\neq$ 6, hashed-id 21 $\neq$ 22; August named-arm is high; this is not a generation-variance CI. English entities leave parametric cues. Named-arm opaque two-path keeps those names and is still mostly UNKNOWN (24/32, 151/200), not a dump of the headquarters city. 2Wiki-CF Wikipedia leak is 75/200 and OpenAI overlap leak 50/200 (Appendix G). Both-opaque unique is 200/200 with names sealed. HMAC two-path rates vary by generator (6/32, 15/32, 12/32). The paper’s central distinction between selection and traversal is inferred from the answer node rather than measured: we never score an emitted (r_1, r_2) . Named-arm selective accuracy stays high; Grok no-ARM opaque commits are near chance (Section 5.2). Header $\times$ decoy at $n=200$ is Table 4; Table 9 is the $n=32$ discovery. K2 and no-ARM were not run at 512. MOVIE-A32 vs. MOVIE-A100 (17/32 vs. 29/32) is a thinner ARM on WikiMovies; $n=200$ Condition A uses the Wiki-H5 topology ARM, not that banner. Dual novel OpenAI UNKNOWN includes empty completions. Forced-choice, scored (r_1, r_2) , and an HQ-city swap are not in this evaluation. The ARM line was not randomized independently of case-id wording on the original lock. Composer hashed-id H5-cyclic used `--preamble none`; the named-arm Composer lock used the Cursor preamble, so that sign reversal is uninterpretable.

Ethical considerations

Public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the WikiMovies/MetaQA graph as released (Miller et al., 2016; Zhang et al., 2018). Appendix cells use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020), Spider (CC-BY-SA-4.0; Yu et al., 2018), and FinQA as released (Chen et al., 2021). No personally sensitive information. Replies are hash-pinned, not a training dump. OIR is a diagnostic, not a confidentiality mechanism (English names remain on the primary 2×2).

8 Conclusion

Unique-route sealed execution succeeds with the start in q (opaque unique OpenAI 32/32 and 200/200; both-opaque unique 32/32 on three families). On the city-typed $n=200$ freeze, named-arm opaque two-path is 38/200 (UNKNOWN 151, decoy 11); hashed case ids at 512 tokens are 134/200 (one NO_OUTPUT); alias $k=3$ is 38/46/43 vs. 134/130/127. Exact-match under two sealed routes is strongly sensitive to prompt surface: case-id wording moves 96 golds at 512 tokens; at 4096/medium, H5 vs. no-ARM on constant decoys is McNemar 4 vs. 111 (Table 4); hashed-id no-ARM English vs. opaque is 197/200 vs. 174/200. Grok no-ARM opaque commits are near chance, so “UNKNOWN rather than decoy” is not a three-family law.

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A Protocol, scoring, and models

Quizzes (runs/) and locks (results/): <https://github.com/pankajjnits/oir> (Python ≥ 3.10 ; hash-pin results/SHA256SUMS, not HEAD). OpenAI: gpt-5.6-sol, no tools, temperature omitted. Named-arm and 512-token hashed-id omit reasoning_effort. Header×decoy at 4096 sets reasoning_effort medium. August stores the parsed answer, not

model/usage; $k=3$ sidecars store returned_model. Composer/Grok: Cursor agent SDK and a reply-only preamble OpenAI never received. Cite Dual free-form as *ff. Empty Dual/no-hop completions were stored as UNKNOWN; Wiki-H5 and Wiki-OO were not. We never impute n.r. as zeros.

Wiki-H5 seed 20260821, 16-byte keys from oir-fact-2x2-iso-v1; 32 of 73 CEO rows; decoys $i+1 \bmod 32$ (Wilson slightly optimistic). Wiki-H5 $n=32$ isolation is 32×5 named arms; $n=32$ header $\times$ decoy is 224 queries per family; $n=200$ header $\times$ decoy at 4096/medium is 1400; gpt56n200h512 queried only OPQ_H5_CYC. Wiki-OO uses oir-entrel-2x2-iso-v1; CEO-NL seed 20260820. The $n=12/n=24$ three-arm is **batch** (one shared key oir-ceiling-three-arm-v1); isolation $n=32$ is per-item keys. Wiki-H5-200 is a separate freeze (wikidata_ceo_hops_n200.json, seed 20260915; $P31 \in \{Q515, Q1549591, Q1093829, Q1637706, Q3957, Q5119, Q13539802, Q5153359, Q7930989\}$; SPARQL verify 200/200). HMAC oir-fact-2x2-iso-n200-v1; $n=32$ prompts are not rewritten. Tags: gpt56n200; header gpt56n200abl; 512 H5-cyclic gpt56n200h512 (gold 134, UNKNOWN 44, decoy 20, other 1, NO_OUTPUT 1). Alias $k=3$ of named-arm / H5-cyclic: gpt56n200k3r2/r3 (46/200, 43/200) and gpt56n200h512k3r2/r3 (130, 127); they do not replace the headline locks. The $n=200$ shuffle re-queries the freeze under tag gpt56n200 on stem factorial_2x2_iso_shuffle_n200_ (new completions; case ids F200S_ rather than F200_). It is not a second model snapshot and does not overwrite the headline lock. WikiMovies $n=200$ keeps the $n=100$ prefix and adds 100 actors.

Non-city P159 golds: Broadway, NRK_Marienlyst, Fox_Plaza, CNN_Center, Lod_Railway_Station; New_Jersey is a state. OpenAI two-path decoys are items 27 and 31 (both non-city). City slice 5/26 (decoy 0) is underpowered; $n=200$ is decoy 11. An archived Movie-B ARM banner is 25/100; locked topology-only B is 3/100. CEO free English hashes q piecewise.

Normalization and HMAC. Strip; collapse whitespace to _; map other non-alphanumeric characters (except underscore, period, hyphen) to _ using [A-Za-z0-9_.-] (Zürich $\mapsto$ Z_rich; pure CJK keeps Unicode); truncate to 64; empty maps to EMPTY. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. HMAC-SHA256 is over UTF-8 bytes with a per-item 16-byte key; the token is E plus the first 6 digest bytes as 12 hex digits. Truncation is not injective over $\{0, 1\}^*$; 48-bit width is instance-wise injective on these small atom sets. Entity and relation types are not domain-separated. On each scored Wikidata 2 $\times$ 2 instance the normalized entity strings and relation strings are disjoint, and no HMAC collision among prompt atoms was observed. Sealing is deterministic given the instance key; the same plaintext need not map to the same token across prompts.

Composer/Grok named-arm ($n=32$). Table 7 is the SDK roster, not a ranking of families. UNKNOWN is an explicit line (missing/unparsed 0; both-named 0). Composer “other” is truncated gold Lod on ENG_UNIQUE, OPAQUE_UNIQUE, and plan, plus Los Angeles for gold Fox_Plaza on OPAQUE_UNIQUE.

Wikidata named-arm (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
English, unique	31/0/1/0	29/0/2/1	32/0/0/0
English, two paths	31/0/1/0	26/0/6/0	4/0/28/0
Opaque, unique	32/0/0/0	27/0/3/2	30/0/2/0
Opaque, two paths	6/2/24/0	10/0/22/0	1/0/31/0
Opaque two paths + plan	32/0/0/0	28/0/3/1	32/0/0/0

Table 7: Four-way outcomes: gold / decoy / UNKNOWN / other. Composer 2.5 and Grok 4.5 used Cursor’s agent SDK with a reply-only preamble OpenAI never received. Body Table 3 is the OpenAI lock.

Opaque generator (OpenAI, start in q)	Unique	Two-path
Keyed HMAC (E+12hex)	32/32	6/32
Random permutation (E+12hex)	31/32	15/32
Random alphanumeric (not HMAC-shaped)	29/32	12/32

Table 8: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson: 29/32 $\Rightarrow$ [0.76, 0.97], 12/32 $\Rightarrow$ [0.23, 0.55], 15/32 $\Rightarrow$ [0.31, 0.64]. Alphanumeric two-path is all abstention or gold (decoy 0).

B Generator robustness

A random permutation of HMAC-shaped labels on the same graphs preserves unique > two-path (OpenAI 31/32 vs. 15/32; Composer 28/32 vs. 2/32; Grok 31/32 vs. 0/32). An English-entity ambiguity sweep $k=1..5$ (constant decoy) keeps OpenAI at 29, 29, 28, 27, 28; Composer falls to 11/32; Grok is 0/32 for $k \geq 2$. Disney names remain, so k is not isolated from entity semantics.

Header $\times$ decoy uses hashed case ids that do not name the condition on the same 32 people (header_decoy_ablation_iso_*.json; body Section 5.2). Cells are named by header and decoy: H5 keeps the Wiki-H5 ARM line, K2 keeps a shorter two-hop ARM, and “no ARM” deletes the line; cyclic decoys attach record $i+1$, constant decoys reuse the first eligible pool record ($n=32$: Cogent Communications $\rightarrow$ Washington, DC; $n=200$: San_Jose $\times$ 199, London $\times$ 1). OpenAI 4096/medium $n=32$ golds:

ARM line (OpenAI, hashed ids, 4096/medium)	Cyclic	Constant
no ARM	25/32	27/32
H5	20/32	10/32
K2	30/32	26/32

Table 9: Header $\times$ decoy gold counts on the same 32 people. English, no ARM, cyclic is 29/32. H5 McNemar 14 vs. 4, $p=0.031$; no-ARM gold 25 vs. 27, McNemar 1 vs. 3, $p=0.625$. Body Table 4 is the $n=200$ lock.

Composer/Grok the same day, prompt file verbatim (`--preamble none`): H5 6/2 and 0/0; K2 20/13 and 0/0; no ARM 21/18 and 6/1; English 26 and 24. Named-arm English Grok verbatim (grok45pn) stays 4/32: the Cursor preamble is not that cell. ENG_NONE_CYC 24/32 differs in both ARM and case ids (Grok 24/32 vs. no-ARM opaque 6/32; McNemar 18 vs. 0). Grok no-ARM opaque commits are 6/3 cyclic and 1/5 constant (7 vs. 8 pooled). Constant decoy does not raise the Wiki-H5-header cell on any family.

C Hypothesis status

Hypothesis	Test	Status
H1. Opacity alone blocks traversal	unique-path, start in q	not supported (32/32)
H2. Written hop restores e2e (control)	two-path + plan; SealRouter	control (32/32)
H3. Input-binding control (CEO piecewise q)	CEO isolation NL	control (0/32); not selection
H4. Matched demos restore shown seals	dual-path matched vs. novel	OpenAI 32/32 vs. 0/32
H5. End-to-end accuracy degrades when opaque relations meet path ambiguity	matched 2×2 , start in q	named-arm 25 vs. 0 (city 21 vs. 0); Sec. 5.1
H6. Movie KB, hashed verbs + two-hop ARM (two-factor check)	WikiMovies Cond. B $n=100$	OpenAI two-path 3/100; archived banner 25/100
H7. Readable relations with opaque entities	Wikidata OE vs. OO	supported for OpenAI here
H8. Prompt surface moves two-path exact match	case-id at 512 cyclic; ARM $\times$ decoy at 4096	$n=200$ case-id +96; H5 McNemar 95 vs. 10; Sec. 5.2

Table 10: Hypothesis status. H4 exact-match is OpenAI Dual; Composer free-form novel is decoy 32/32 (composer25ff). H5 is named-arm WIKI-H5 (OpenAI). H6 hashes verbs *and* uses a two-hop ARM; an archived banner is 25/100. H7 is OpenAI-specific (OE vs. WIKI-OO). H8 is prompt surface on the same graphs (Section 5.2).

CEO-NL isolation $n=32$	Composer 2.5	OpenAI	Grok 4.5
Sealed plan / SealRouter	32/32	32/32	32/32
Free sealed English (piecewise q)	0/32	0/32	0/32

Table 11: Missing-start control, not WIKI-H5. Isolation $n=32$. Batch $n=12/n=24$ uses one shared key (Appendix A).

D Dual-path arms

We name the arms by what the demos teach. **Matched-relation** (asymmetric or balanced noise): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced noise): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan**: gold relation sequence is written in the prompt.

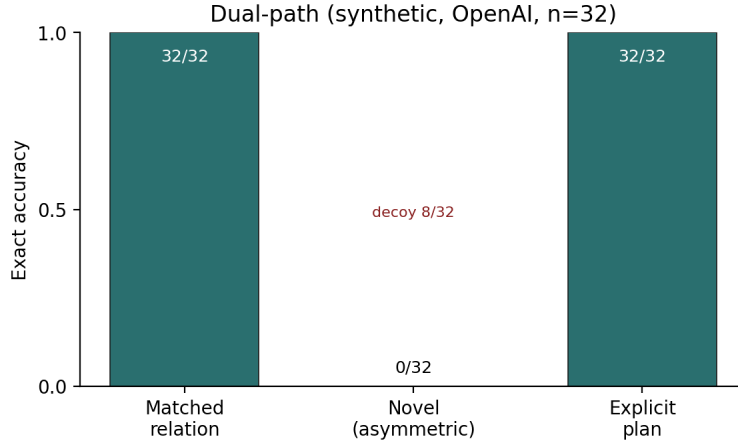


Figure 2: DUAL (synthetic, OpenAI $n=32$): matched 32/32; novel 0/32 (asymmetric decoy 8); explicit plan 32/32.

Isolation $n=32$ is the reported cell. Cite Dual free-form as composer25ff/grok45ff; the isolation-tag Composer helper is not free-form. Local mistral/qwen35 are not in the roster (plan n.r.).

E No-hop binding (JSON)

Arm (no written hop)	OpenAI	Composer 2.5	Grok 4.5
Schema list only	0/32	0/32	0/32
English protocol (no hop)	32/32	32/32	32/32
Same σ , silent	6/32	19/32	31/32
Same σ + equality recipe	32/32	32/32	32/32
JSONKEY : match broken	0/32	0/32	0/32
Sealed Q, two routes, recipe	31/32	25/32	32/32
Sealed Q, two routes, vague hints*	13/32	23/32	31/32

Table 12: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- σ overlap and vague hints are family-dependent, not a three-family restore. OpenAI silent 6/32 and vague-hint 13/32 include 24 and 15 empty completions stored as UNKNOWN. *Composer vague-hint distractor 1/32. JOIN was not evaluated at $n=32$. Wilson: 32/32 $\Rightarrow$ [0.89, 1.00], 0/32 $\Rightarrow$ [0.00, 0.11], 6/32 $\Rightarrow$ [0.09, 0.35], 31/32 $\Rightarrow$ [0.84, 0.99]. Silent same- σ OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

Prompts are fixed English blocks, not WIKI-H5. Equality means token identity (Appendix J). English JSON keys alone do not restore the walk (1/6).

F Sealing middleware

Bind a typed full name to one graph atom, then seal. Isolation without binding is 0/6 vs. layer 6/6 (Composer, OpenAI, unversioned Claude; plaintext in the packed prompt: 0). Those files attach a join: an execution control, not the free WIKI-H5 6/32 cell.

G Packaging probes

Not the matched 2×2 . Free sealed English 0/12 vs. plan 12/12 on table/JSON/vault (Composer, OpenAI). Spider identifier-in- q $n=100$: EQ 51/100, namespace-break 0/100 (OpenAI; not official EM). If the identifier is not in q ($n=26$), the namespace-break cell is 20/26: an English q plus legend can still bind. Excel NL names both predicates: Composer 0/16 vs. plan 16/16. FinQA NL 0/8 vs. gold programs 8/8 (Chen et al., 2021). 2Wiki-CF: Composer plain NL 200/200; OpenAI 150/200 with overlap leak 50/200 (not a 2Wiki eval). Long context: NL 0/8, plans 8/8. Rename under a second HMAC: 12/12. SealRouter and gold-SQL are non-LLM ceilings (Figure 3).

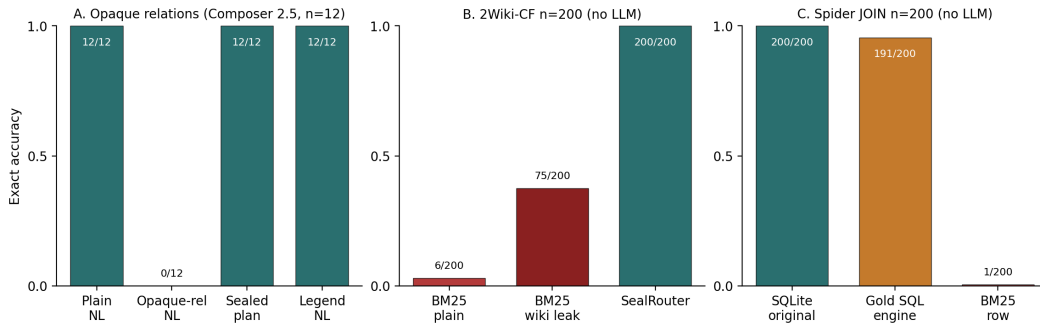


Figure 3: Non-LLM and packaging controls. Left: Composer $n=12$ (plain 12/12, opaque-rel NL 0/12, plan 12/12, legend 12/12). Middle: 2Wiki-CF BM25 6/200, Wikipedia leak 75/200, SealRouter 200/200. Right: Spider sqlite $n=200$ (200/200), gold-SQL 191/200, BM25 1/200. Spider identifier-in- q EQ 51/100 is a different cell (Appendix G).

H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except `[start]`: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That $n=32$ template cell is not MOVIE-B.

I Sign-flip of the H5 interaction

Let $d_i = \mathbf{1}[\text{opaque correct}] - \mathbf{1}[\text{English correct}]$ and $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$. Of 32 WIKI-H5 items, 24 have $c_i = -1$, one has $c_i = -2$, 7 have $c_i = 0$; $T = -26$. Under exchangeability of unique-path and two-path opacity effects, $p = 2^{-24} = 5.96 \times 10^{-8}$. That p coincides numerically with the two-path McNemar p (25 vs. 0; McNemar, 1947); the tests are not the same. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

J Equality recipe (JSON cell)

Verbatim protocol used in Table 12, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”

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